CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in machine learning and NLP, from text data pipelines, model training and fine-tuning to evaluation harnesses and clean, modular Python code. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by a strong foundation in Python development across enterprise and start-up environments.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning a text data pipeline for knowledge extraction end-to-end, from document ingestion and chunking to prompt engineering and indexing (agentic, deterministic, and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Raised extraction precision/recall from 33%/45% to 93%/90% by evaluating each iteration against an SAP-system-derived ground truth, while designing a multi-agent LangGraph system (entity-extraction, relationship-analysis, consolidation, and reflection agents in parallel) for regulatory and technical documents.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Integrated the knowledge graph into Claude Code through a Model Context Protocol (MCP) query service with millisecond-latency retrieval, and built Expert Interface, a graph curation web app (React frontend, Python microservices backend) on Azure where domain experts validate and approve changes.
- **Tech stack: Python, LangChain, LangGraph, LiteLLM, RAG, GraphRAG, Vector Search, MCP, Docker, Kubernetes, Azure, FastAPI, React, Neo4j, MongoDB, Postgres**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows, advised on where agentic AI could support them, and translated the requirements into a solution approach.
- Built and deployed an agentic AI application with LangGraph that routes requests across vector database queries, general inference, and summarization, with a React frontend and a Python FastAPI backend tested with Postman.
- Containerized the stack with Docker, deployed it to AWS with CDK, and version-controlled it in GitLab with CI/CD.
- **Tech stack: Python, LangChain, LangGraph, FastAPI, React, Postman, Docker, GitLab CI/CD, AWS (CDK, EC2, Lambda)**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Built an automated, reproducible evaluation harness for the PrismaChat RAG system with RAGAS and PytrecEval, scoring generation and retrieval metrics against chemist-provided ground truth.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data, and built retrieval pipelines covering chunking strategies, vector databases, indexing, and fine-tuning models for better retrieval with PyTorch and Hugging Face Transformers.
- Set up AWS infrastructure (Lambda, SQS, DynamoDB) with CDK for the evaluation workflow.
- **Tech stack: Python, PyTorch, Hugging Face Transformers, RAGAS, PytrecEval, LangChain, Azure AI Search, AWS (CDK, Lambda, SQS, DynamoDB)**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (viscosity, reaction progression) from time-series process data, and benchmarked them against empirical curve-fitting.
- Engineered lag features and rolling windows from sensor data for the tree models; fed raw sequences to the LSTM.
- Built a Streamlit dashboard where chemists simulate parameter changes before running physical batches.
- **Tech stack: Python, TensorFlow/Keras, Scikit-Learn, XGBoost, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, TensorFlow, Scikit-Learn**

PROJECTS

Project: Enhancing Image Classification | CNN | RNN | ResNet
- Applied CNN, RNN, and ResNet architectures to permuted and unpermuted MNIST and Fashion-MNIST, achieving 93% accuracy.

Project: Meme Generation using LLMs | Open Llama 3B | QLoRA
- Fine-tuned Open Llama 3B with QLoRA to generate responses to educational meme-related prompts.

TECHNICAL SKILLSET

- **Programming:** Python
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow/Keras, Scikit-Learn, XGBoost, LSTM, CNN, time-series analysis (ARIMA, SARIMA, Prophet), fine-tuning (QLoRA)
- **NLP:** Hugging Face Transformers, NLTK, information extraction, LangChain
- **Data Analysis:** Pandas, NumPy, SciPy, Matplotlib, Streamlit, SQL
- **Databases:** Postgres, MongoDB, Neo4j, DynamoDB, vector databases (Qdrant, Pinecone, Azure AI Search)
- **Generative AI:** LangChain, LangGraph, LiteLLM, RAG, GraphRAG, agents and multi-agent systems, MCP, prompt engineering, RAGAS, PytrecEval
- **Cloud & Software Architecture:** AWS (CDK, Lambda, EC2, SQS, S3), Azure, Docker, Kubernetes, FastAPI (REST), Python microservices, React
- **Development Process:** Git, GitLab CI/CD, Postman, Agile, Claude Code

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering, Minor in Computer Science & Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - The Complete Neural Networks Boot camp
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT
- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)